Model Card: Adult Income Classification

1. Model Details

Logistic Regression trained on the UCI Adult Income dataset. Tools: scikit-learn, IBM AIF360 and SHAP.

2. Intended Use

Educational analysis of classification, fairness and explainability. Not intended for real hiring, lending, insurance or eligibility decisions.

3. Data and Factors

Records after cleaning: 30,162. Target: income above $50K. Gender was excluded from predictive inputs and retained for auditing. The dataset reflects historical US census-related income patterns.

4. Performance

Baseline accuracy: 0.8467. Reweighted accuracy: 0.8400.

5. Fairness Evaluation

Baseline positive prediction rates: female 0.0920; male 0.2563. Baseline disparate impact: 0.3588. After Reweighing: 0.6341.

6. Explainability

SHAP LinearExplainer was used to explain feature contributions. The summary plot is saved as outputs/shap_summary.png.

7. Limitations and Ethical Considerations

The dataset is not representative of India. Historical income labels may reflect structural inequalities. Removing gender does not prevent proxy discrimination. The 0.80 disparate-impact threshold is a screening guideline, not proof of discrimination. Reweighing does not guarantee fairness across every metric or group.

8. Recommendations

Before deployment, conduct independent subgroup audits, assess data quality and predictive performance, document limitations, and monitor for changing outcomes. Do not use this educational model for consequential decisions.